

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts of image formation, pinhole to perspective camera models, homogeneous coordinates, and multi-view geometry and interpret real-world visual scenes using OpenCV or MATLAB. (BL2)

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks:20

Annexure A Concept Mapping

Code of Conduct:

I Hashini S / 192524035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Hashini S

RUBRICS FOR CALCULATION

Concept Mapping						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured.	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

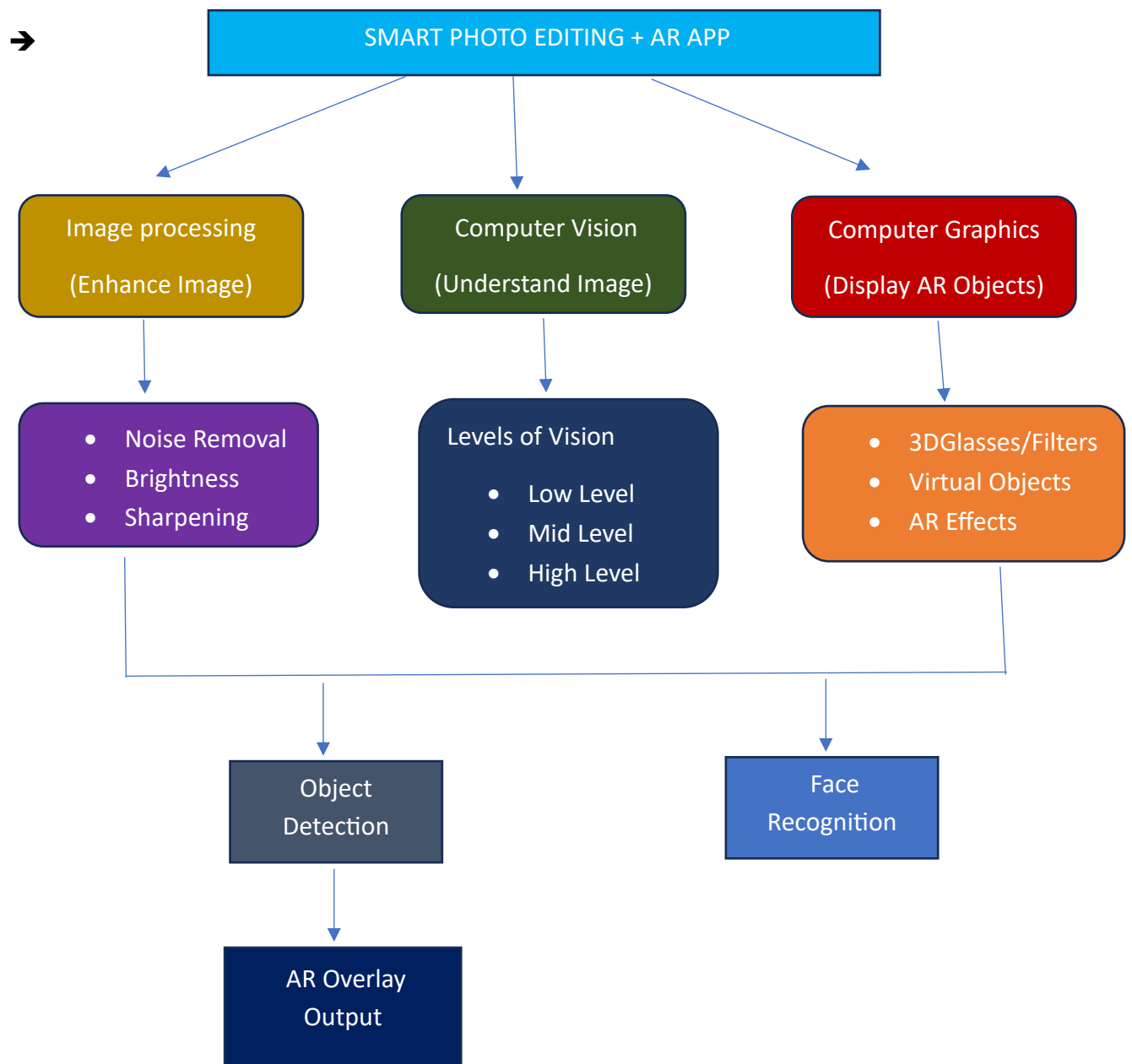
1. A company is developing a smart photo editing + object recognition + AR app.

Create a concept map connecting:

- Image Processing
- Computer Vision
- Computer Graphics
- Levels of Vision
- Applications

Show:

- Workflow of the system
- Interconnections between components
- Real-world examples



Real-World Examples with Explanation

Application	Explanation
Google Photos	Uses image processing to automatically enhance brightness, colors, and remove noise.
Google Lens	Uses computer vision to recognize objects, text, plants, and landmarks.
Snapchat Filters	Detects the face using computer vision and adds 3D filters using computer graphics .
Instagram AR Filters	Recognizes facial features and overlays virtual effects in real time.
Pokémon GO	Uses computer vision to understand the environment and computer graphics to place 3D Pokémon in the real world.
IKEA Place App	Detects room surfaces and places virtual furniture using augmented reality.

Explanation

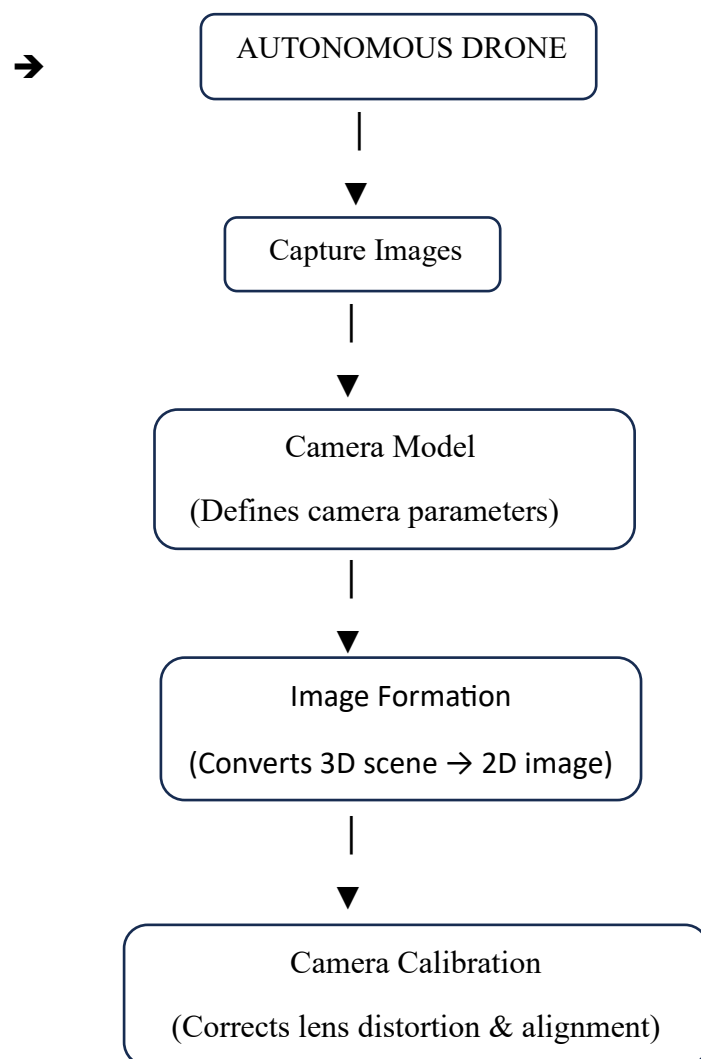
1. **Image Processing** enhances the captured image by removing noise, adjusting brightness, and sharpening it.
2. **Computer Vision** analyzes the processed image to detect and recognize objects or faces.
3. **Levels of Vision** work in three stages:
 - **Low Level:** Detects edges and basic features.
 - **Mid Level:** Extracts shapes and important features.
 - **High Level:** Identifies and understands objects.
4. **Computer Graphics** adds virtual objects, 3D models, stickers, or AR effects to the recognized image.
5. **Workflow:** Image Capture → Image Processing → Computer Vision → Levels of Vision → Object/Face Recognition → Computer Graphics → AR Output.
6. **Real-world Applications:** Snapchat and Instagram use face detection for AR filters, Google Lens recognizes objects, and Pokémon GO places virtual objects in the real world.

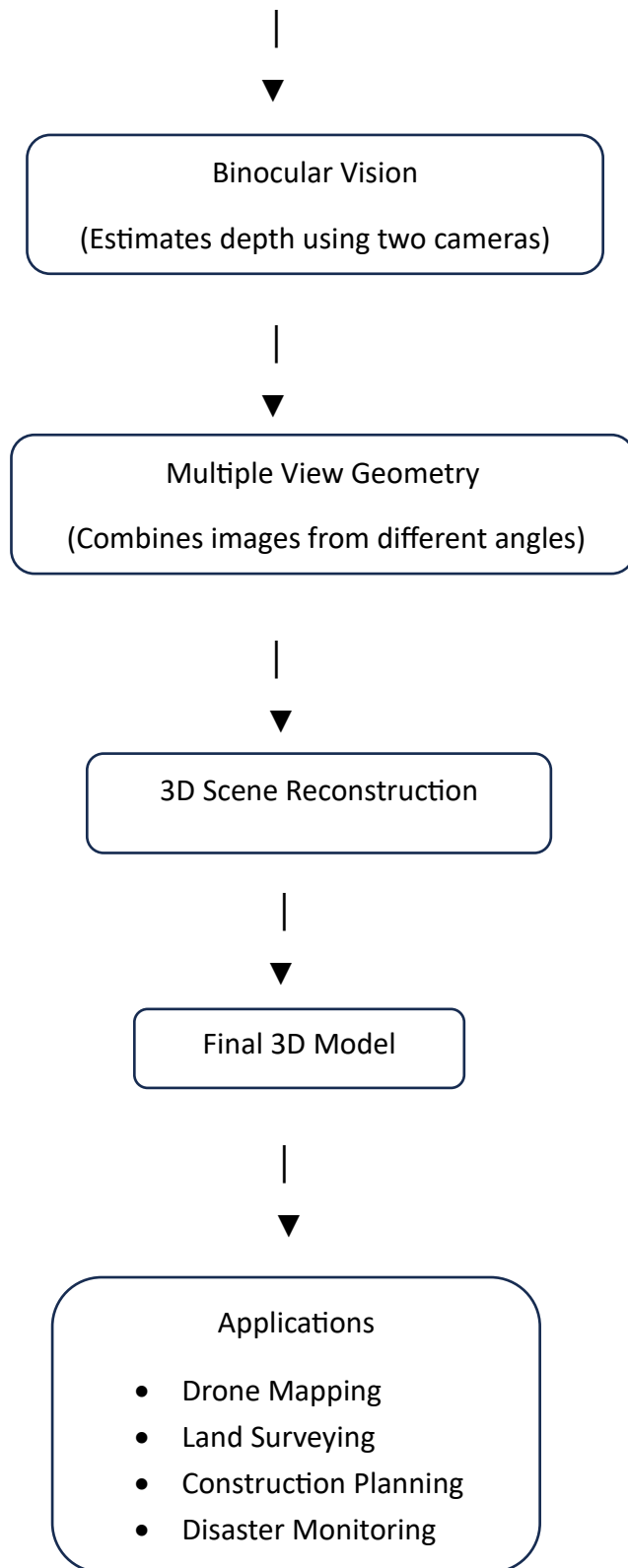
2. An autonomous drone captures images, calibrates cameras, estimates depth, and reconstructs a 3D scene. Draw a concept map linking:

- Camera Model
- Image Formation
- Camera Calibration
- Binocular Vision
- Multiple View Geometry

Include:

- Flow of data
- Dependencies
- Output (3D reconstruction)





Dependency Explanation:

- **Camera Model** defines how the camera captures images.
 - **Image Formation** creates a 2D image from the 3D scene.
 - **Camera Calibration** corrects distortion for accurate measurements.
 - **Binocular Vision** estimates the depth of objects.
 - **Multiple View Geometry** combines multiple images to build a complete 3D scene.
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3. Output (3D Reconstruction)

The final output is a **3D model** of the environment, which can be used for:

- Drone mapping
 - Land surveying
 - Construction planning
 - Disaster monitoring
 - Archaeological site reconstruction
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4. Simple Explanation

1. The drone captures images using its cameras.
2. The **camera model** defines how the camera records the scene.
3. **Image formation** converts the real 3D world into 2D images.
4. **Camera calibration** removes lens distortion and aligns the cameras correctly.
5. **Binocular vision** and **multiple view geometry** estimate depth and combine images from different angles.
6. The final output is an accurate **3D reconstruction** of the scene for mapping and navigation.